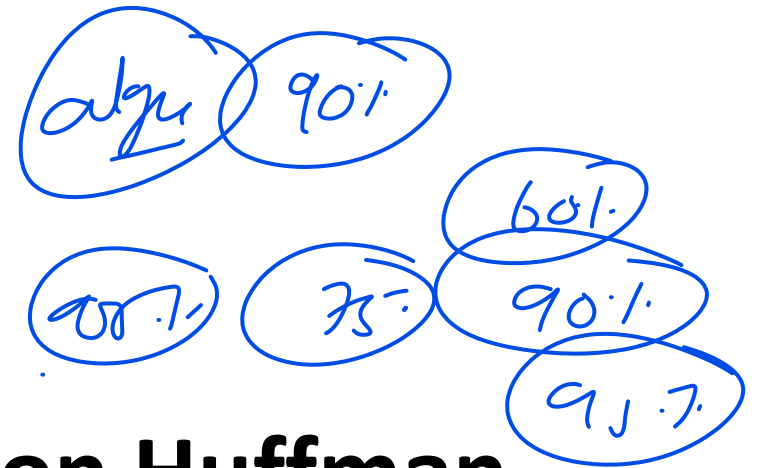
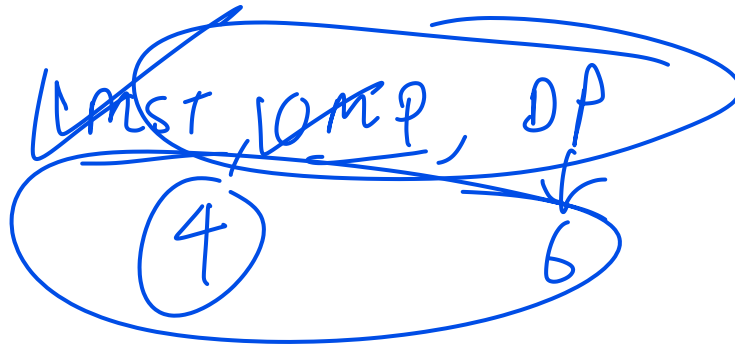




Lec_62

Practice/GATE Questions on Huffman Coding

GATE | GATE-CS-2017 (Set 2) | Question 58

A message is made up entirely of characters from the set $X = \{P, Q, R, S, T\}$. The table of

Character	Probability
P	0.22
Q	0.34
R	0.17
S	0.19
T	0.08
Total	1.00

probabilities of each character is shown below :

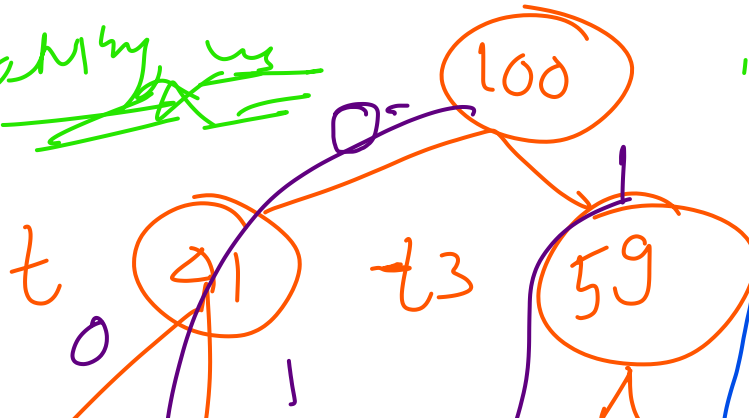
message of 100 characters over X is encoded using Huffman coding. Then the expected length of the encoded message in bits is _____

- ☒ 225
- ☐ 226
- ☐ 227
- ☐ 228

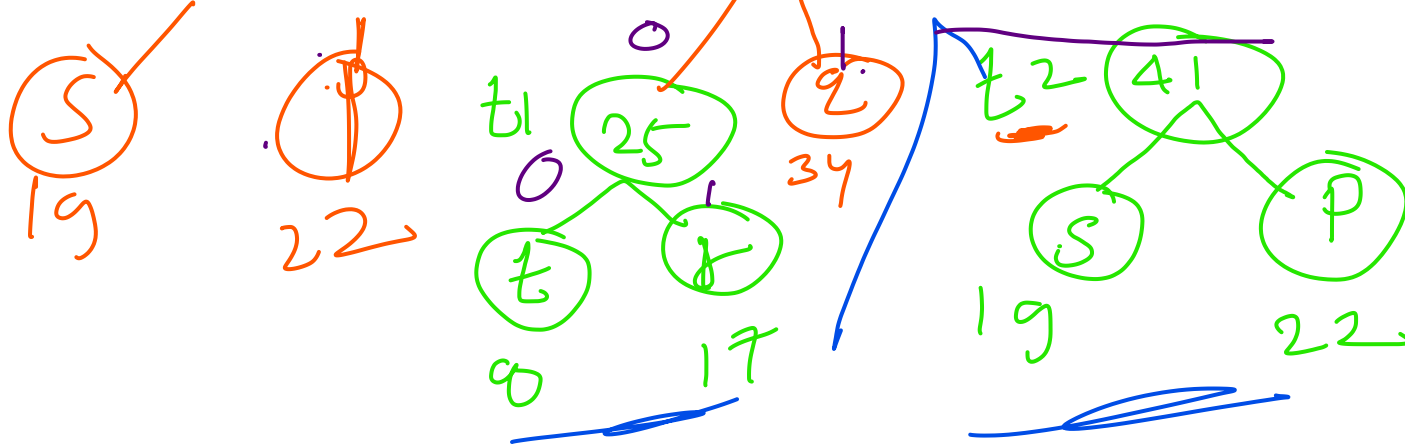
~~$p = 22$~~ , ~~$q = 34$~~ , ~~$r = 17$~~ , ~~$s = 19$~~ , ~~$t = 8$~~ , ~~$t_1 = 25$~~

$t_2 = 41$, $t_3 = 59$

same as



[In Between distributed small trees cause]



$p = 01$, level 4
 $q = 11$, 2
 $r = 101$, 3
 $s = 00$, 2
 $t = 100$, 3

$$p = 22, \quad q = 34, \quad r = 17, \quad s = 19, \quad t = 8$$

$$\begin{aligned}
 \text{Total work} &= 22(2) + 34(2) + 17(3) + 19(2) + 8(3) \\
 &= 44 + 68 + 51 + 38 + 24 \\
 &= 112 + 89 + 24 \\
 &= 112 + 113 \\
 &= 225
 \end{aligned}$$

GATE | GATE-CS-2007 | Question 76

Suppose the letters a, b, c, d, e, f have probabilities $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$, $\frac{1}{16}$, $\frac{1}{32}$, $\frac{1}{32}$ respectively. Which of the following is the Huffman code for the letter a, b, c, d, e, f?

☐ 0, 10, 110, 1110, 11110, 11111

☐ 11, 10, 011, 010, 001, 000

☐ 11, 10, 01, 001, 0001, 0000

☐ 110, 100, 010, 000, 001, 111

$$a = \frac{1}{2}, b = \frac{1}{4}, c = \frac{1}{8}, d = \frac{1}{16}$$

$$e = \frac{1}{32}, f = \frac{1}{32}$$

but assume we are sending 32 char

$$\text{So, } a = 32 \times \frac{1}{2} = 16$$

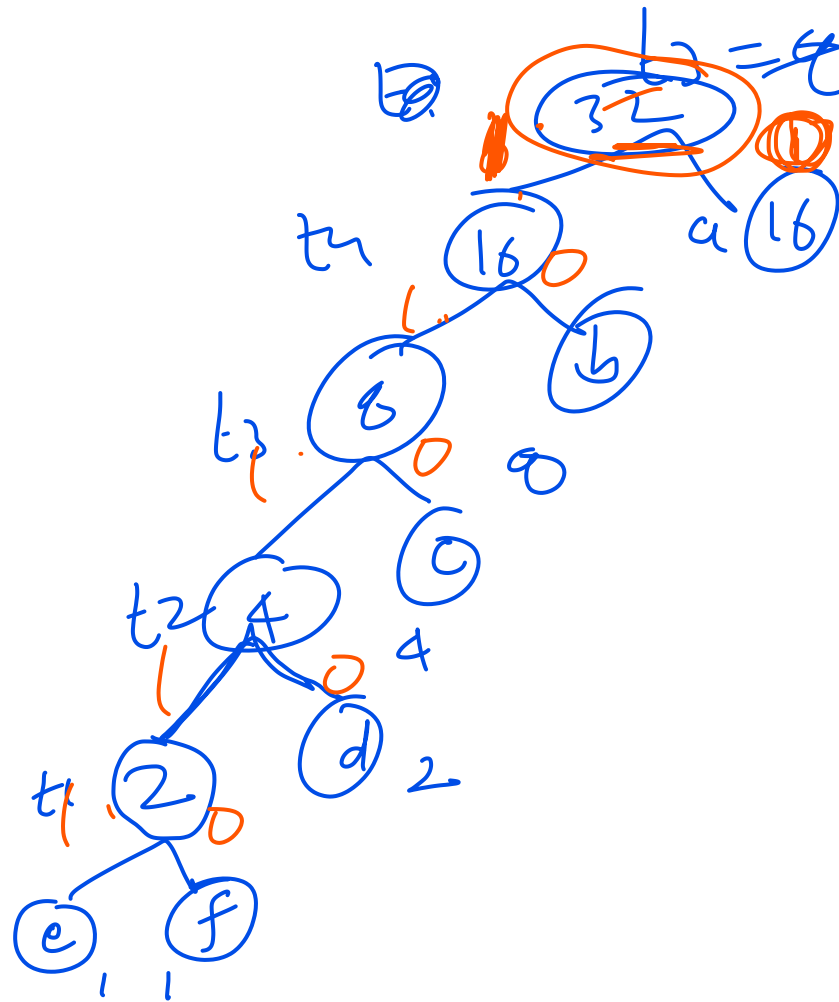
$$b = \frac{1}{4} \times 32 = 8$$

$$c = 32 \times \frac{1}{8} = 4$$

$$d = \frac{1}{16} \times 32 = 2$$

$$e = f = \frac{1}{32} \times 32 = 1$$

$a=16, b=8, c=4, d=2, e=1, f=1$ $t_1=2, t_2=4$



$t_1=16$

$a=0$

$b=10$

$c=110$

$d=1110$

$e=11110$

$f=11111$

Leaf Char.

1	16
2	8
3	4
4	2
5	1
5	1

Total Char = 32

Total bit = $16(1) + 8(2) + 4(3) + 4(2) + 5(1) + 5(1)$

$= 16 + 16 + 12 + 8 + 10$

$= 32 + 30$
 $= 62$

T.B = 62

T.C = 32

$62/32$

$\Rightarrow 1.9$

$\frac{31}{16}$

GATE | GATE-CS-2007 | Question 77

Suppose the letters a, b, c, d, e, f have probabilities $1/2$, $1/4$, $1/8$, $1/16$, $1/32$, $1/32$ respectively. What is the average length of Huffman codes?

☐ 3

☐ 2.1875

☐ 2.25

☒ 1.9375

3.) $m_1=0.9$, $m_2=0.95$, $m_3=0.3$, $m_4=1.1$, $m_5=0.5$

If message received 10101110001101011001100 what is actual message in Huffman coding

~~$m_1=0.9$~~

~~$m_2=0.95$~~

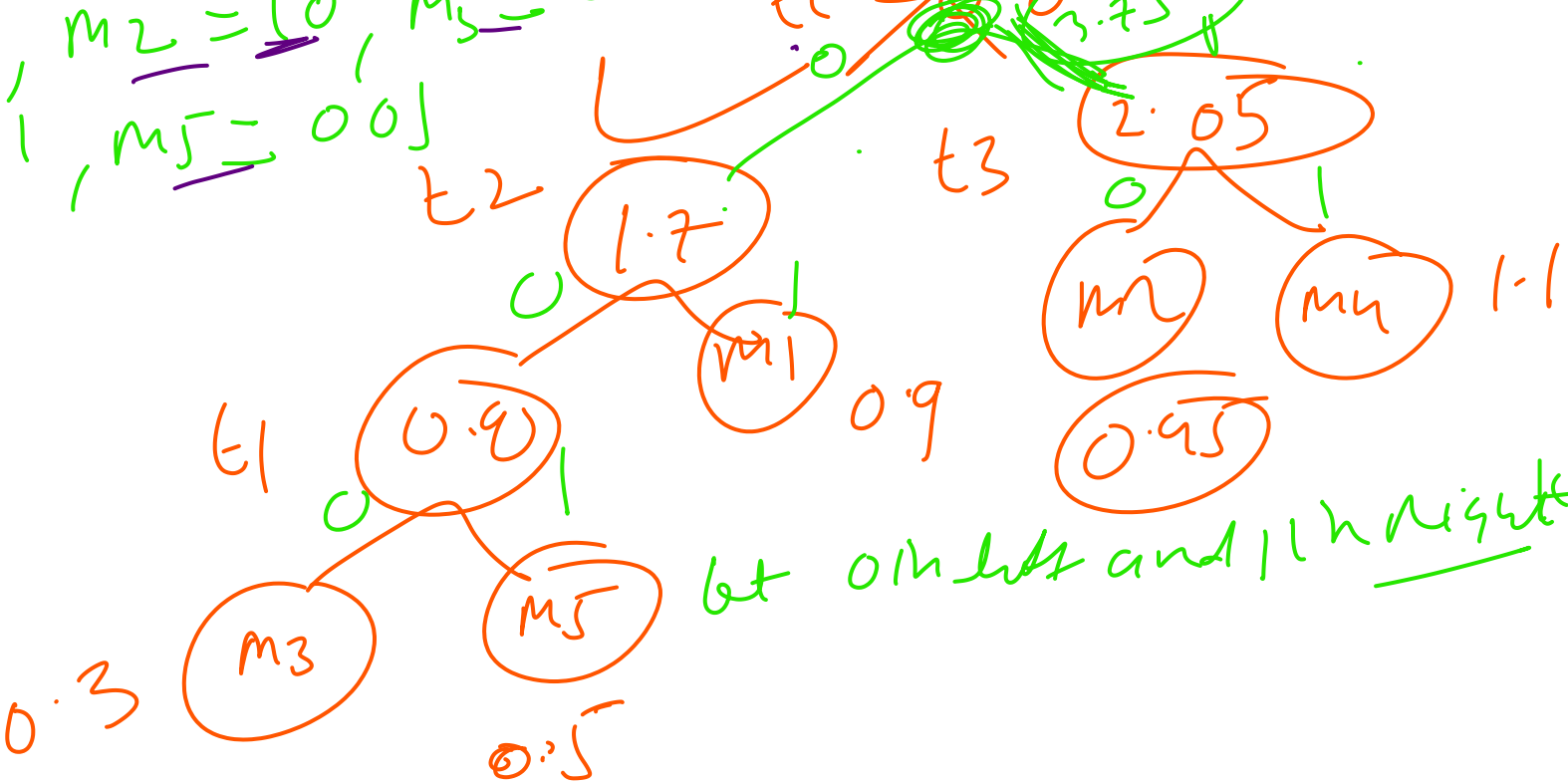
~~$m_3=0.3$~~

~~$m_4=1.1$~~

~~$m_5=0.5$~~

$m_1=01$, $m_2=10$, $m_3=000$
 $m_4=11$, $m_5=001$

~~$t_1=5.9$~~ , ~~$t_2=1.7$~~ , ~~$t_3=2.05$~~



④ optm. (my)

